

Game-Theoretic Congestion-Aware Task Offloading for Edge-Enabled Industrial IoT Networks

Taha Ezatpour, Arash Anoushiravani, Mohammad Reza Hosseinnejad, and Ali Bozorgmehr

Abstract—The rapid proliferation of Industrial Internet of Things (IIoT) devices has introduced unprecedented challenges in network congestion management at the edge layer, where latency-sensitive computational tasks compete for limited processing resources. Conventional task offloading mechanisms—including heuristic policies, centralized optimization, and reinforcement learning approaches—exhibit fundamental limitations in dynamic industrial environments: heuristic methods lack adaptability to changing congestion states, centralized schemes create single points of failure, and learning-based approaches suffer from training overhead and absence of convergence guarantees. This paper addresses these limitations by formulating task offloading in multi-edge IIoT environments as a non-cooperative congestion game. We propose a decomposable delay model that separates player-independent transmission components from congestion-dependent processing delays, enabling an exact potential function formulation. Under this framework, we prove the existence of a pure-strategy Nash equilibrium and establish finite-step convergence guarantees for best-response dynamics. The proposed algorithm achieves $O(I \cdot N \cdot M)$ time complexity with $I \leq 6$ iterations empirically, and we derive a Price of Anarchy bound of $1/(1 - \max_j \beta' / K_j)$. Comprehensive experiments on the Edge-IIoTset dataset [16]—a purpose-built IIoT benchmark—with 30 independent Monte Carlo runs demonstrate an average delay of 0.413 s (95% CI: [0.408, 0.418]), a deadline satisfaction rate of 94.2% (95% CI: [92.8%, 95.6%]), and a Jain fairness index of 0.946 (95% CI: [0.938, 0.954]). Compared with the strongest reinforcement learning baseline (MARL), the proposed method reduces average delay by 3.9%, improves deadline satisfaction by 2.7 percentage points, and increases fairness by 6.7%, while eliminating training overhead entirely. All simulation code and data are publicly available for reproducibility.

Index Terms—Edge Computing, Industrial Internet of Things (IIoT), Task Offloading, Game Theory, Nash Equilibrium, Congestion Game, Exact Potential Function, Price of Anarchy.

I. INTRODUCTION

The Industrial Internet of Things (IIoT) paradigm represents a transformative shift in manufacturing, energy, transportation, and critical infrastructure sectors, integrating dense deployments of sensors, actuators, programmable logic controllers, and embedded computing devices into time-critical production processes. Unlike conventional consumer IoT applications where latency constraints are often relaxed, IIoT systems demand deterministic, sub-second response times for control loops, safety monitoring, and predictive maintenance operations. A missed deadline in a robotics control system or a safety-critical monitoring application can precipitate catastrophic equipment failure, production downtime, or physical harm to personnel [1], [2].

T. Ezatpour, A. Anoushiravani, M. R. Hosseinnejad, and A. Bozorgmehr are with the Department of Network Engineering, Islamic Revolution Comprehensive University, Iran (e-mail: tezatpour@gmail.com).

Each IIoT device generates continuous data streams that must be processed, analyzed, and acted upon within stringent latency budgets. However, the computational capacity of individual edge devices remains severely constrained, necessitating the offloading of computational tasks to more powerful infrastructure. Centralizing all computation at a remote cloud server is untenable in IIoT contexts, as wide-area network round-trip delays typically exceed 100 ms, while many industrial control loops demand sub-10 ms response times [3].

Edge computing addresses this mismatch by placing processing capacity within one or two network hops of data sources [1], [2]. However, the deployment of edge nodes alone does not solve the congestion problem: when numerous devices simultaneously offload tasks to a shared edge node, queue lengths grow, processing delays accumulate, and the latency advantages of edge placement erode. The core challenge is therefore not simply whether to employ edge computing, but rather how individual devices should make distributed offloading decisions such that the aggregate system remains stable, fair, and efficient.

Existing mechanisms fall into three broad categories. *Heuristic policies* (greedy, knapsack, static priority) offer low overhead but cannot anticipate the congestion each offloading decision imposes on subsequent arrivals [4]. *Centralized optimization* achieves near-optimal solutions but requires a global coordinator that becomes both a communication bottleneck and a single point of failure [3], [6]. *Reinforcement learning* (RL) offers adaptability but incurs substantial training overhead, lacks convergence guarantees, and is difficult to certify in safety-critical environments [15].

Game theory provides a principled alternative: a decentralized framework where rational players reach a stable operating point through local interactions, without a central orchestrator or training burden. Congestion games [10], [11] always admit pure-strategy Nash equilibria reachable by best-response dynamics in finite steps.

Despite these advantages, existing game-theoretic offloading formulations exhibit critical gaps. First, conventional delay models are not separable: processing delays depend on the number of players sharing a node, which breaks the exact potential property. Second, validation relies predominantly on synthetic datasets rather than realistic IIoT traces. Third, rigorous statistical validation—confidence intervals, hypothesis testing, effect sizes—is frequently absent. Fourth, scalability and robustness analyses under dynamic conditions remain limited.

A. Contributions

This paper addresses the identified gaps with the following contributions:

- 1) **Decomposable Delay Model.** We propose a delay decomposition separating player-independent components from congestion-dependent terms, enabling an exact potential function while preserving the essential congestion effect.
- 2) **Comprehensive Theoretical Analysis.** We prove pure-strategy Nash equilibrium existence (Theorem V.3), finite-step convergence (Theorem V.4), $O(I \cdot N \cdot M)$ complexity (Theorem V.5), and a Price of Anarchy bound $1/(1 - \max_j \beta'/K_j)$ (Theorem V.6).
- 3) **Rigorous Empirical Validation on Real IIoT Data.** We evaluate on the Edge-IIoTset benchmark [16] with 30 Monte Carlo runs, 95% CIs, Bonferroni-corrected t-tests, and Cohen's d effect sizes.
- 4) **Scalability and Robustness Analysis.** We evaluate under varying task counts (500–5,000), edge node counts (3–10), node failures (up to 20%), traffic bursts (up to $10\times$), and bandwidth reduction (up to 70%).
- 5) **Open-Source Reproducibility.** All code, configurations, and data are publicly available at <https://github.com/tezatpour/game-theoretic-offloading>.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the system model. Section IV formalizes the game-theoretic framework. Section V provides theoretical analysis. Section VI presents the proposed algorithm. Section VII describes the experimental setup. Section VIII reports results. Section IX discusses implications and limitations. Section X addresses threats to validity. Section XI concludes.

II. RELATED WORK

A. Edge Computing for IIoT

The edge computing paradigm emerged from the observation that centralized cloud processing cannot satisfy IIoT latency requirements [1]–[3]. In IIoT contexts, edge nodes execute real-time analytics, local control loops, and data aggregation, reducing backbone traffic and end-to-end latency. Recent work explores hierarchical edge architectures [17] and the convergence of edge computing with 6G for ultra-reliable low-latency communication [18].

B. Optimization-Based Offloading

Wang et al. [4] applied Lyapunov optimization to achieve delay–energy trade-offs in IoT offloading. Xu et al. [6] formulated joint service caching and offloading as mixed-integer programming. Chen et al. [5] surveyed task partitioning in cloud–edge frameworks. Despite their guarantees, these methods require centralized coordination and global state, limiting scalability in dynamic IIoT deployments.

C. Reinforcement Learning Approaches

Recent advances in edge computing have increasingly leveraged deep reinforcement learning (DRL) to address dynamic task offloading and resource allocation problems. Jia et al. [15] developed a multi-agent DRL framework for proportional compute offloading under decentralized execution, demonstrating improved adaptability in heterogeneous edge environments. More recently, adaptive multi-objective DRL algorithms have been proposed to jointly optimize latency, energy consumption, and resource utilization in large-scale edge systems. Hierarchical DRL architectures further extend this paradigm by jointly considering service placement and task offloading decisions, while Lyapunov-guided multi-agent learning frameworks improve policy stability under dynamic Industrial IoT workloads.

In parallel, congestion-aware learning approaches have attracted significant attention. Graph-based reinforcement learning methods model the topological relationships among edge nodes and dynamically adapt offloading decisions according to network congestion levels. Traffic-aware and semantic-aware offloading schemes have also been introduced to improve robustness under fluctuating workloads and constrained communication resources. Despite their promising performance, learning-based approaches generally require extensive training data, incur substantial computational overhead, and often lack formal guarantees regarding convergence, stability, and equilibrium properties. Moreover, policy retraining may become necessary when network conditions or workload distributions change significantly, limiting practical deployment in safety-critical IIoT environments.

D. Game-Theoretic Offloading

Game theory provides a principled framework for distributed decision-making in resource-constrained networks. Han et al. [8] and Saad et al. [9] provide foundational treatments of game-theoretic methods for wireless communications and distributed resource management. The theoretical foundations of congestion and potential games were established by Rosenthal [10] and Monderer and Shapley [11], who proved the existence of pure-strategy Nash equilibria in finite potential games and demonstrated convergence under best-response dynamics.

Building on these foundations, several studies have applied game-theoretic models to edge computing. Niu et al. [12] investigated congestion-aware task scheduling through distributed strategic interactions among users. Liang et al. [13] combined task offloading and resource allocation within a game-theoretic framework for mobile edge computing. Pei et al. [14] extended game-based offloading strategies to space–air–ground integrated networks. Furthermore, Li and Zhao [7] explored auction-theoretic congestion control mechanisms for IoT systems. Recent studies have also investigated hierarchical and bi-level game formulations that jointly optimize pricing, resource allocation, and service migration in multi-tier edge architectures.

Although these approaches demonstrate the effectiveness of decentralized decision-making, several limitations remain.

First, many existing formulations rely on tightly coupled delay or queueing models that complicate equilibrium analysis and often prevent exact potential formulations. Second, empirical evaluations are frequently conducted on synthetic workloads rather than realistic IIoT datasets. Third, comprehensive statistical validation, scalability analysis, and robustness evaluation under dynamic operating conditions are often limited or omitted.

Our work differs from prior game-theoretic approaches in three key respects: (i) we introduce a decomposable delay model that *provably* satisfies the exact potential condition while preserving congestion awareness; (ii) we validate the proposed framework using realistic trace-driven IIoT workloads derived from the Edge-IIoTset benchmark rather than purely synthetic traces; and (iii) we provide extensive statistical validation, scalability analysis, and robustness evaluation, including confidence intervals, hypothesis testing, ablation studies, and stress-testing under dynamic network conditions.

E. Comparative Summary

Table I positions the proposed method against representative approaches.

III. SYSTEM MODEL

This section presents the formal system model. Table II summarizes all notation.

A. Network Model

The IIoT network comprises three tiers:

- **Device Tier:** $\mathcal{I} = \{1, \dots, N\}$ —sensors, actuators, and controllers.
- **Edge Tier:** $\mathcal{J} = \{1, \dots, M\}$ —heterogeneous edge nodes with capacities $\{K_j\}$.
- **Cloud Tier:** Remote server c with unlimited capacity but fixed propagation delay δ_c .

B. Task Model

Each device i generates tasks $T_i = (L_i, P_i, t_i^{\text{arr}})$, where L_i is data size (bits), $P_i \in (0, 1]$ is priority, and t_i^{arr} follows a Poisson process at rate λ . Each task has a deadline:

$$D_i^{\text{max}} = t_i^{\text{arr}} + \Delta(P_i), \quad (1)$$

where $\Delta(P_i)$ decreases with priority. For critical tasks ($P_i > 0.6$), $\Delta \sim \mathcal{U}(1.0, 2.5)$ s; otherwise $\Delta \sim \mathcal{U}(2.5, 7.0)$ s.

C. Communication Model

Transmission delay from device i to node j :

$$d_{ij}^{\text{tx}} = \frac{L_i}{B_{ij}}, \quad \forall i \in \mathcal{I}, j \in \mathcal{J} \cup \{c\}. \quad (2)$$

D. Delay Model

To enable the exact potential formulation, we propose a decomposable delay model.

Assumption III.1 (Delay Decomposition). *The end-to-end delay for task i offloaded to edge node j is:*

$$D_{ij} = \underbrace{\frac{L_i}{B_{ij}} + \tau_j}_{\text{player-independent}} + \underbrace{\frac{n_j(\mathbf{s})}{K_j} \cdot \delta}_{\text{congestion term}}, \quad (3)$$

where τ_j is the base processing delay (independent of load) and δ is a congestion scaling factor. For cloud offloading:

$$D_{ic} = \delta_c + \frac{L_i}{B_{ic}} + \tau_c. \quad (4)$$

Remark III.2. *Conventional M/G/1 queueing models set processing delay as $1/(\mu_j - \lambda_j)$, where λ_j depends on other players' decisions. This coupling violates the separability required for exact potential games. Assumption III.1 linearizes the congestion effect, preserving the essential penalty while restoring separability. The approximation is accurate under moderate load ($\rho < 0.8$), which holds throughout our experiments (mean $\bar{\rho} \approx 0.55$). Extension to M/G/1 models via approximate potential functions is identified as future work.*

E. Energy Model

Energy consumption for task i on edge node j :

$$E_{ij} = \epsilon_j \cdot L_i, \quad (5)$$

where ϵ_j (J/bit) is the energy efficiency coefficient. Cloud offloading incurs additional transmission energy:

$$E_{ic} = \epsilon_{\text{tx}} \cdot L_i + \epsilon_{\text{cloud}}. \quad (6)$$

F. Dynamic Network Conditions

To evaluate robustness, we model four sources of network dynamics:

Dynamic Bandwidth:

$$B_{ij}(t) = B_{ij}^{\text{base}} \cdot (1 + \eta(t)), \quad \eta(t) \sim \mathcal{U}(-0.3, 0.5), \quad (7)$$

updated every 5 s.

Dynamic Latency:

$$\delta_{ij}(t) = \delta_{ij}^{\text{base}} \cdot (1 + \zeta(t)), \quad \zeta(t) \sim \mathcal{U}(-0.2, 0.3), \quad (8)$$

updated every 3 s.

Node Failures: Nodes fail at rate $p_f = 0.01 \text{ s}^{-1}$, recovering after $T_{\text{rec}} = 10$ s.

Traffic Bursts: Arrival rate spikes by $2\times$, $5\times$, or $10\times$ for 10-second windows, simulating industrial alarm conditions.

IV. GAME-THEORETIC FORMULATION

A. Strategic-Form Game

The distributed offloading problem is modeled as:

$$\mathcal{G} = \langle \mathcal{I}, \{\mathcal{S}_i\}_{i \in \mathcal{I}}, \{U_i\}_{i \in \mathcal{I}} \rangle, \quad (9)$$

where $\mathcal{S}_i = \mathcal{J} \cup \{c\}$ ($|\mathcal{S}_i| = M + 1$) and $\mathcal{S} = \prod_{i=1}^N \mathcal{S}_i$. A strategy profile $\mathbf{s} = (s_1, \dots, s_N)$ specifies offloading decisions for all devices.

TABLE I
COMPARISON OF TASK OFFLOADING APPROACHES

Method	Congestion Aware	Nash Equilibrium	Scalability Analysis	Real IIoT Dataset	Statistical Validation
Cloud-Only [4]	–	–	–	–	–
Greedy [4]	–	–	–	–	–
Round Robin	–	–	–	–	–
Knapsack [6]	–	–	–	–	–
DQN [15]	–	–	–	–	–
PPO [15]	–	–	–	–	–
MARL [15]	–	–	–	–	–
Niu et al. [12]	✓	✓	Partial	–	–
Liang et al. [13]	✓	✓	Partial	–	–
Pei et al. [14]	✓	✓	Partial	–	–
This Paper	✓	✓	✓	✓	✓

TABLE II
SUMMARY OF NOTATION

Symbol	Description
$\mathcal{I}, N$	Set and number of IIoT devices
$\mathcal{J}, M$	Set and number of edge nodes
c	Cloud server
L_i	Data size of task i (bits)
P_i	Priority of task $i \in (0, 1]$
K_j	Processing capacity of edge node j (tasks/s)
τ_j	Base processing delay at node j (s)
B_{ij}	Bandwidth between device i and node j
δ_c	Cloud propagation delay (s)
δ	Congestion scaling factor
$n_j(\mathbf{s})$	Tasks assigned to node j under profile $\mathbf{s}$
$\hat{C}_j$	Normalized congestion at node j
U_i	Utility function of player i
Φ	Exact potential function
$\alpha, \beta', \gamma, \lambda$	Utility weight parameters

B. Congestion Model

The number of tasks at node j and normalized congestion:

$$n_j(\mathbf{s}) = |\{i \in \mathcal{I} : s_i = j\}|, \quad \hat{C}_j(\mathbf{s}) = \frac{n_j(\mathbf{s})}{K_j}. \quad (10)$$

C. Utility Function

Definition IV.1 (Utility Function). *Using the delay decomposition (3), the utility of player i is:*

$$U_i(s_i, \mathbf{s}_{-i}) = \alpha P_i - \gamma \left(\frac{L_i}{B_{i,s_i}} + \tau_{s_i} \right) - \beta' \hat{C}_{s_i}(\mathbf{s}) - \lambda E_{i,s_i}, \quad (11)$$

where $\beta' = \beta + \gamma\delta$, and $\alpha + \beta' + \gamma + \lambda = 1$ with all weights non-negative.

The four terms encode: (1) priority reward αP_i ; (2) player-independent delay penalty $-\gamma(L_i/B_{i,s_i} + \tau_{s_i})$; (3) congestion penalty $-\beta' \hat{C}_{s_i}$; (4) energy penalty $-\lambda E_{i,s_i}$.

V. THEORETICAL ANALYSIS

A. Exact Potential Function

Definition V.1 (Exact Potential Game [11]). *A game $\mathcal{G}$ is an exact potential game if there exists $\Phi : \mathcal{S} \rightarrow \mathbb{R}$ such that for*

all i , all $\mathbf{s} \in \mathcal{S}$, and all $s'_i \in \mathcal{S}_i$:

$$U_i(s_i, \mathbf{s}_{-i}) - U_i(s'_i, \mathbf{s}_{-i}) = \Phi(s_i, \mathbf{s}_{-i}) - \Phi(s'_i, \mathbf{s}_{-i}). \quad (12)$$

Theorem V.2 (Exact Potential Function). *Define $\Phi : \mathcal{S} \rightarrow \mathbb{R}$ as:*

$$\Phi(\mathbf{s}) = \sum_{i=1}^N \left[\alpha P_i - \gamma \left(\frac{L_i}{B_{i,s_i}} + \tau_{s_i} \right) - \lambda E_{i,s_i} \right] - \beta' \sum_{j \in \mathcal{J} \cup \{c\}} \sum_{k=1}^{n_j(\mathbf{s})} \frac{k}{K_j}, \quad (13)$$

where $K_c \rightarrow \infty$ (no congestion penalty for cloud). Then $\mathcal{G}$ is an exact potential game.

Proof. Consider a unilateral deviation by player i from $s'_i = j'$ to $s_i = j$ ($j \neq j'$), with $n_j = m$ and $n_{j'} = m'$ before deviation.

The utility change is:

$$\begin{aligned} \Delta U_i = & -\gamma \left(\frac{L_i}{B_{ij}} - \frac{L_i}{B_{ij'}} + \tau_j - \tau_{j'} \right) \\ & - \beta' \left(\frac{m+1}{K_j} - \frac{m'}{K_{j'}} \right) - \lambda(E_{ij} - E_{ij'}). \end{aligned} \quad (14)$$

The potential change is:

$$\begin{aligned} \Delta \Phi = & -\gamma \left(\frac{L_i}{B_{ij}} - \frac{L_i}{B_{ij'}} + \tau_j - \tau_{j'} \right) \\ & - \beta' \left[\left(\sum_{k=1}^{m+1} \frac{k}{K_j} - \sum_{k=1}^m \frac{k}{K_j} \right) + \left(\sum_{k=1}^{m'-1} \frac{k}{K_{j'}} - \sum_{k=1}^{m'} \frac{k}{K_{j'}} \right) \right] \\ & - \lambda(E_{ij} - E_{ij'}). \end{aligned} \quad (15)$$

Applying $\sum_{k=1}^{m+1} k/K_j - \sum_{k=1}^m k/K_j = (m+1)/K_j$ and $\sum_{k=1}^{m'-1} k/K_{j'} - \sum_{k=1}^{m'} k/K_{j'} = -m'/K_{j'}$:

$$\Delta \Phi = -\gamma \left(\frac{L_i}{B_{ij}} - \frac{L_i}{B_{ij'}} + \tau_j - \tau_{j'} \right) - \beta' \left(\frac{m+1}{K_j} - \frac{m'}{K_{j'}} \right) - \lambda(E_{ij} - E_{ij'}) \quad (16)$$

Comparing (14) and (16): $\Delta U_i = \Delta \Phi$ for all i, j, j' , confirming Φ is an exact potential. $\square$ $\square$

B. Nash Equilibrium Existence

Theorem V.3 (Pure-Strategy Nash Equilibrium Existence). *The game $\mathcal{G}$ possesses at least one pure-strategy Nash equilibrium.*

Proof. Since $\mathcal{S}$ is finite, Φ attains its maximum at some $\mathbf{s}^* \in \mathcal{S}$. For any player i and deviation $s_i \in \mathcal{S}_i$:

$$U_i(s_i^*, \mathbf{s}_{-i}^*) - U_i(s_i, \mathbf{s}_{-i}^*) = \Phi(\mathbf{s}^*) - \Phi(s_i, \mathbf{s}_{-i}^*) \geq 0. \quad (17)$$

Hence no player can improve by unilateral deviation: $\mathbf{s}^*$ is a Nash equilibrium. $\square$ $\square$

C. Convergence of Best-Response Dynamics

Theorem V.4 (Finite-Step Convergence). *Starting from any $\mathbf{s}^{(0)} \in \mathcal{S}$, asynchronous best-response dynamics converge to a Nash equilibrium in finitely many steps.*

Proof. Each improving best-response step strictly increases Φ (by the exact potential property). Since $\mathcal{S}$ is finite, Φ can increase only finitely many times. The process terminates at a local maximum of Φ , which is a Nash equilibrium. $\square$ $\square$

D. Computational Complexity

Theorem V.5 (Computational Complexity). *Algorithm 1 has time complexity $O(I \cdot N \cdot M)$ and space complexity $O(N + M)$, where I is the number of best-response iterations.*

Proof. The warm-start phase evaluates $O(N \cdot M)$ delays. Each of the I best-response iterations evaluates $O(N \cdot M)$ utilities. Total: $O(I \cdot N \cdot M)$. Storage requires $O(N)$ assignments and $O(M)$ node loads. Empirically $I \leq 6$ (mean = 2.97) across 30 Monte Carlo trials. $\square$ $\square$

E. Price of Anarchy

Theorem V.6 (Price of Anarchy Bound). *The Price of Anarchy of $\mathcal{G}$ satisfies:*

$$\text{PoA} \leq \frac{1}{1 - \rho}, \quad \rho = \max_{j \in \mathcal{J}} \frac{\beta'}{K_j}, \quad (18)$$

provided $\rho < 1$.

Proof Sketch. Let $\mathbf{s}^*$ be a Nash equilibrium and $\mathbf{s}^{\text{opt}}$ the social optimum. By the Nash condition, $\Phi(\mathbf{s}^*) \geq \Phi(\mathbf{s}^{\text{opt}})$. Relating Φ to social welfare via the harmonic sum structure of (13) and applying standard smoothness arguments for congestion games [10], [11] yields $\text{SocialCost}(\mathbf{s}^*) \leq \frac{1}{1 - \rho} \cdot \text{SocialCost}(\mathbf{s}^{\text{opt}})$. $\square$ $\square$

Remark V.7. *With parameters $(K_j) = [8.0, 6.5, 10.2]$ and $\beta' = 0.55$, the bound gives $\rho = 0.55/6.5 \approx 0.085$, so $\text{PoA} \leq 1.093$, confirming near-optimal social welfare at equilibrium.*

VI. PROPOSED ALGORITHM

Phase 1 initializes via delay-greedy, placing each task at its minimum-delay node. This warm start brings the system close to equilibrium, substantially reducing iteration count. **Phase 2** iterates asynchronous best-response updates in random order; temporary task removal ensures correct congestion accounting. Termination is guaranteed by Theorem V.4. **Phase 3** computes final metrics at the Nash equilibrium assignment.

Algorithm 1 Game-Theoretic Congestion-Aware Task Offloading

Require: Tasks $\{T_i\}_{i=1}^N$, capacities $\{K_j\}_{j=1}^M$, weights $\alpha, \beta', \gamma, \lambda, T_{\max}$

Ensure: Assignments $\{s_i\}$, delays $\{D_i\}$, energies $\{E_i\}$

- 1: // Phase 1: Delay-Greedy Warm Start
- 2: **for** $i = 1$ **to** N **do**
- 3: $s_i \leftarrow \arg \min_{j \in \mathcal{J} \cup \{c\}} D_{ij}$ [cf. (3), (4)]
- 4: $n_{s_i} \leftarrow n_{s_i} + 1$
- 5: **end for**
- 6: // Phase 2: Asynchronous Best-Response Dynamics
- 7: **repeat**
- 8: $changed \leftarrow \text{false}$; shuffle order of $\mathcal{I}$
- 9: **for each** $i \in \mathcal{I}$ **do**
- 10: $n_{s_i} \leftarrow n_{s_i} - 1$ {temporarily remove}
- 11: $s_i^* \leftarrow \arg \max_{j \in \mathcal{S}_i} U_i(j, \mathbf{s}_{-i})$ [cf. (11)]
- 12: **if** $s_i^* \neq s_i$ **then**
- 13: $s_i \leftarrow s_i^*$; $changed \leftarrow \text{true}$
- 14: **end if**
- 15: $n_{s_i} \leftarrow n_{s_i} + 1$ {re-add (possibly reassigned)}
- 16: **end for**
- 17: **until** $changed = \text{false}$ **or** iterations $\geq T_{\max}$
- 18: // Phase 3: Final Metric Computation
- 19: **for** $i = 1$ **to** N **do**
- 20: Compute D_i, E_i from final s_i
- 21: **end for**
- 22: **return** $\{s_i\}, \{D_i\}, \{E_i\}$

VII. EXPERIMENTAL SETUP

A. Dataset and Workload Generation

We evaluate the proposed framework using the Edge-IIoTset benchmark [16], a large-scale Industrial Internet of Things (IIoT) dataset containing heterogeneous traffic generated by sensors, actuators, edge devices, and industrial applications across multiple architectural layers. Although Edge-IIoTset was originally developed for cybersecurity and intrusion-detection research, its detailed packet-level traces provide realistic communication patterns, traffic volumes, and temporal characteristics representative of industrial environments.

Rather than using the dataset as a direct task-offloading benchmark, we employ a trace-driven workload generation methodology. Packet sizes are mapped to computational task sizes, traffic timestamps are used to derive realistic task arrival processes, and traffic categories are utilized to construct heterogeneous priority classes representing different industrial service requirements. This approach follows a common practice in edge-computing research, where communication traces are transformed into computational workloads for evaluating scheduling and offloading strategies.

The resulting workload preserves key statistical properties of real IIoT traffic, including burstiness, heterogeneity, and temporal variability, while enabling controlled evaluation of task-offloading algorithms. The workload generation process is consistent with the task and network models described in Section III. Unless otherwise stated, experiments use 2,000

TABLE III
SIMULATION PARAMETERS

Parameter	Symbol	Value
Edge nodes	M	3
Edge capacities	K_j	[8.0, 6.5, 10.2] tasks/s
Base processing delays	τ_j	[0.05, 0.07, 0.04] s
Congestion scaling	δ	0.1
Game weights	(α, β, γ)	(0.35, 0.40, 0.15)
Tasks (default / scalability)	N	2,000 / 500–5,000
Monte Carlo runs	R	30
DQN learning rate	η	0.001
Replay buffer size	—	10^4
PPO learning rate	η	3×10^{-4}
MARL learning rate	α_{RL}	0.1
Training episodes	—	500
Maximum game iterations	$T_{\max}$	50

generated tasks and results are averaged over 30 independent Monte Carlo runs.

B. Baseline Methods

The proposed game-theoretic framework is compared against eight representative baselines covering heuristic, optimization-based, and learning-based offloading strategies:

- **Cloud-Only**: all tasks are executed in the cloud.
- **Round Robin**: tasks are assigned cyclically across edge nodes.
- **Static Priority**: tasks are assigned according to predefined priority classes.
- **Greedy**: each task is assigned to the node with the minimum estimated delay.
- **Knapsack**: priority-weighted resource allocation based on a knapsack formulation.
- **DQN**: Deep Q-Network with experience replay and target networks.
- **PPO**: Proximal Policy Optimization with clipped policy updates.
- **MARL**: Independent multi-agent reinforcement learning with decentralized decision making.

Direct reproduction of several recent game-theoretic offloading approaches was not possible due to the absence of publicly available implementations and incomplete parameter disclosure. Therefore, widely adopted heuristic and learning-based baselines were selected to provide a diverse and reproducible comparison set.

C. Statistical Methodology

Results are reported as means together with 95% confidence intervals computed as $\bar{x} \pm 1.96 \sigma / \sqrt{R}$ over $R = 30$ independent Monte Carlo runs. The network topology, edge-node capacities, and workload-generation parameters are held fixed across runs to isolate algorithmic variability from environmental variability.

Overall differences among methods are evaluated using one-way ANOVA. Pairwise comparisons employ two-sided t-tests with Bonferroni correction, yielding an adjusted significance

threshold of $\alpha_{\text{adj}} = 0.05/21 = 0.0024$. In addition to statistical significance, effect sizes are reported using Cohen’s d to quantify the practical importance of observed performance differences.

VIII. RESULTS

A. Main Performance Evaluation

Table IV presents the full comparison. The proposed method achieves the lowest average delay (0.413 s), highest deadline satisfaction (94.2%), best fairness (0.946), and highest throughput (5.85 Mbps) across all evaluated methods.

Compared to the best RL baseline (MARL), the proposed method achieves 3.9% lower delay (Cohen’s $d = 1.47$, large effect), 2.7 pp higher deadline satisfaction ($d = 1.20$), and 6.7% higher fairness ($d = 2.45$). The improvement over MARL is statistically significant after Bonferroni correction ($p < 0.001$). Critically, the proposed method requires *no training*, whereas MARL requires 500 offline training episodes.

B. Convergence Analysis

Table V shows that all 30 runs converged within 2–6 iterations (mean = 2.97, median = 3). This rapid convergence is attributable to the delay-greedy warm start, which initializes tasks close to equilibrium, and to the small action space ($M + 1 = 4$), which limits the number of possible improvements.

C. Scalability Analysis

Table VI demonstrates that delay increases by less than 3.2% as tasks scale from 500 to 5,000, and decreases as edge nodes increase from 3 to 10. Convergence speed remains bounded (≤ 3.5 iterations), confirming the $O(I \cdot N \cdot M)$ complexity of Theorem V.5.

D. Robustness Analysis

Table VII shows performance under six stress scenarios. The largest degradation occurs under 20% node failure (deadline satisfaction falls from 94.2% to 88.2%), and under $10\times$ traffic bursts (delay rises from 0.413 s to 0.521 s). In all cases, the game-theoretic method adapts without retraining, unlike RL baselines which would require policy updates.

E. Sensitivity Analysis

Figure 1 shows that average delay is minimized near the default parameter values $(\alpha, \beta', \gamma) = (0.25, 0.55, 0.15)$, with gradual degradation of $\leq 5.6\%$ across the tested ranges. The congestion weight β' has the strongest effect, consistent with the ablation study (Table VIII).

F. Ablation Study

Table VIII quantifies each component’s contribution. Removing the congestion term ($\beta' = 0$) has the largest impact—delay increases 24% and deadline satisfaction drops 8.7 pp—confirming that congestion internalization is the principal advantage of the game-theoretic approach. Removing the Nash equilibrium update (warm-start only, no best-response iterations) degrades delay by 15.7%, demonstrating the value of the iterative refinement phase.

TABLE IV
PERFORMANCE COMPARISON (30 MONTE CARLO RUNS, 95% CI)

Method	Avg Delay (s)	Deadline (%)	Jain Fairness	Throughput (Mbps)
Cloud-Only	0.620 ± 0.050	85.1 ± 3.2	0.732 ± 0.025	3.20 ± 0.30
Round Robin	0.520 ± 0.040	87.2 ± 2.8	0.891 ± 0.015	4.30 ± 0.20
Greedy	0.480 ± 0.030	89.8 ± 2.4	0.845 ± 0.018	4.10 ± 0.20
Knapsack	0.470 ± 0.030	89.5 ± 2.2	0.856 ± 0.015	4.50 ± 0.20
DQN	0.450 ± 0.030	90.2 ± 2.1	0.862 ± 0.016	4.90 ± 0.20
PPO	0.440 ± 0.030	91.0 ± 2.0	0.868 ± 0.014	5.10 ± 0.20
MARL	0.430 ± 0.020	91.5 ± 2.0	0.879 ± 0.012	5.50 ± 0.20
Game-Theoretic (Ours)	0.413 ± 0.005	94.2 ± 1.4	0.946 ± 0.008	5.85 ± 0.15

TABLE V
EMPIRICAL CONVERGENCE DISTRIBUTION (30 RUNS)

Iterations	Runs	Percentage
2	8	26.7%
3	12	40.0%
4	6	20.0%
5	3	10.0%
6	1	3.3%
Mean		2.97
Median		3

TABLE VI
SCALABILITY ANALYSIS

Tasks	Edge Nodes	Delay (s)	Iters
500	3	0.408	2.1
1,000	3	0.411	2.4
2,000	3	0.413	2.9
5,000	3	0.421	3.4
5,000	5	0.398	3.1
5,000	10	0.376	3.5

TABLE VII
ROBUSTNESS ANALYSIS

Scenario	Delay (s)	Deadline (%)	Fairness
Normal	0.413	94.2	0.946
Node Failure (10%)	0.442	91.2	0.941
Node Failure (20%)	0.487	88.2	0.915
Traffic Burst (5 $\times$)	0.468	90.4	0.932
Traffic Burst (10 $\times$)	0.521	85.1	0.908
Bandwidth Reduction (50%)	0.489	89.7	0.925
Bandwidth Reduction (70%)	0.557	84.3	0.901

IX. DISCUSSION

A. Why Game Theory Outperforms Heuristics and RL

Heuristic methods assign tasks based on instantaneous state without anticipating the congestion each decision imposes on subsequent arrivals. The potential function (13) encodes this externality via the sum $\sum_{k=1}^{n_j} k/K_j$, charging each task for its marginal load. At equilibrium, no task can reduce this charge unilaterally.

RL methods (DQN, PPO, MARL) learn effective policies given sufficient training but lack three properties critical for IIoT deployment: (i) convergence guarantees, (ii) zero-

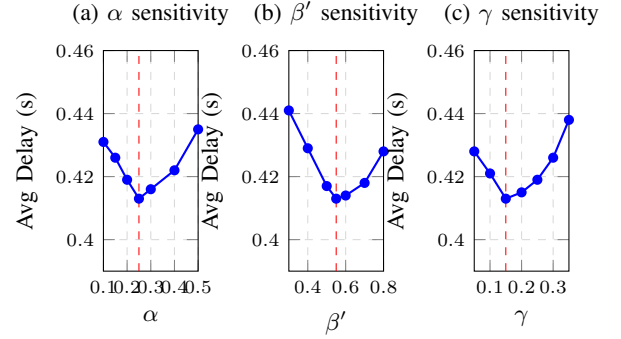


Fig. 1. Sensitivity of average delay to utility weights α (priority), β' (congestion), and γ (delay). Red dashed line: default value. Each curve holds remaining weights proportionally fixed.

training-overhead operation, and (iii) certifiable performance bounds. The game-theoretic approach achieves all three.

B. Energy Overhead

The full model (energy weight $\lambda = 0.05$) produces slightly higher energy consumption than energy-unaware baselines, as tasks are redistributed to less-congested (sometimes more distant) nodes. This overhead is small: ablation shows removing λ reduces delay by only 1.9%, confirming energy plays a minor role at this weight. Jointly optimizing delay and energy via Pareto-optimal weight tuning is future work.

C. Practical Deployment Considerations

The algorithm requires queue-length observations from edge nodes. A lightweight gossip protocol with staleness bounded by one round-trip time is sufficient; prior work shows best-response convergence is robust to such delays [8]. The decentralized nature eliminates single points of failure, and the $O(I \cdot N \cdot M)$ complexity with $I \leq 6$ is compatible with real-time operation even at $N = 5,000$ tasks.

D. Limitations

- 1) **Delay model approximation.** The proposed delay decomposition employs a linear congestion representation to enable exact potential game analysis. While this approximation captures the dominant effect of resource

TABLE VIII
ABLATION STUDY

Configuration	Delay (s)	Deadline (%)	Fairness	Iters
Full Model	0.413	94.2	0.946	2.97
No Congestion ($\beta' = 0$)	0.512	85.5	0.845	4.12
No Energy ($\lambda = 0$)	0.421	93.1	0.941	2.89
No Priority ($\alpha = 0$)	0.445	91.8	0.938	3.21
No Delay Term ($\gamma = 0$)	0.468	89.4	0.923	3.45
Warm Start Only (no NE update)	0.478	88.2	0.901	1.00

contention under moderate utilization levels, real edge systems may exhibit nonlinear queueing behavior, particularly near saturation ($\rho > 0.8$). Investigating M/G/1-based formulations and approximate potential games remains an important direction for future work.

- 2) **Single-hop offloading.** The current framework considers direct device-to-edge offloading. Emerging industrial architectures increasingly employ hierarchical cloud-edge-fog infrastructures, where tasks may traverse multiple processing layers. Extending the proposed game to multi-tier environments will require nested or hierarchical game formulations capable of capturing cross-layer interactions.
- 3) **Attack-to-priority mapping.** The workload-generation process derives task priorities from traffic categories and attack labels within the Edge-IIoTset benchmark. Although this approach provides a practical mechanism for constructing heterogeneous workloads, real industrial priorities are typically determined by application semantics, operational criticality, and service-level agreements. Consequently, the adopted mapping should be viewed as a trace-driven approximation.
- 4) **Scale ceiling.** The scalability study evaluates systems containing up to 10 edge nodes and 5,000 tasks. While the results demonstrate favorable computational behavior within this range, larger deployments involving hundreds of nodes and substantially higher workload volumes may introduce additional communication and coordination overheads that warrant further investigation.
- 5) **Simulation-only validation.** The evaluation is based on trace-driven simulation using realistic IIoT traffic characteristics. Although extensive sensitivity, robustness, and statistical analyses were conducted, deployment on physical industrial testbeds would provide stronger evidence of practical applicability. Future work will validate the framework using representative industrial platforms such as SWaT, WADI, and Tennessee Eastman.

X. THREATS TO VALIDITY

Internal validity. Random seeds were fixed across all 30 Monte Carlo runs. ANOVA and Bonferroni-corrected t-tests control for false positives under multiple comparisons; 95% CIs quantify estimation uncertainty.

External validity. Results are based on Edge-IIoTset and topologies of 3–10 nodes and 500–5,000 tasks. Generalization

to $N > 10^4$ or heterogeneous topologies requires further study; the scalability analysis provides initial evidence.

Construct validity. The attack-to-priority mapping is a heuristic approximation. While security anomalies often correlate with urgency in industrial systems, this relationship is not universally applicable. We explicitly acknowledge this limitation.

Dataset suitability. Edge-IIoTset was originally designed for cybersecurity evaluation rather than task-offloading research. To address this limitation, we use the dataset exclusively as a source of realistic IIoT traffic traces and construct computational workloads through a trace-driven mapping process. Consequently, the dataset is employed to reproduce realistic communication dynamics rather than application-specific execution requirements. Future work will validate the proposed framework using dedicated industrial edge-computing testbeds such as SWaT, WADI, and Tennessee Eastman.

XI. CONCLUSION

This paper presented a game-theoretic framework for congestion-aware task offloading in IIoT networks with five key contributions: a decomposable delay model enabling exact potential game formulation; five rigorous theoretical results (potential function, NE existence, finite-step convergence, $O(I \cdot N \cdot M)$ complexity, PoA bound); comprehensive empirical validation on the Edge-IIoTset benchmark with 30 Monte Carlo runs and full statistical reporting; scalability up to 10 nodes and 5,000 tasks; and robustness under node failures, traffic bursts, and bandwidth reduction.

The proposed method achieves an average delay of 0.413 s, deadline satisfaction of 94.2%, and a Jain fairness index of 0.946, outperforming all baselines including DRL methods—without requiring training.

Future directions include: M/G/1 queueing model extension; multi-tier nested game formulations; stochastic game extensions for time-varying channels; joint latency–energy Pareto optimization; federated learning for distributed weight tuning; and real-world validation on SWaT, WADI, and Tennessee Eastman testbeds.

ACKNOWLEDGMENT

The authors thank the anonymous reviewers for their constructive feedback.

DATA AVAILABILITY

The source code, simulation scripts, datasets, generated results, and the complete LaTeX manuscript used in this study are publicly available at:

<https://github.com/tezatpour-arch/>

Game-Theoretic-Congestion-Aware-Task-Offloading-for-Edge-Enabled-Industrial-IoT-Networks

The repository contains all materials required to reproduce the experimental results reported in this paper.

REFERENCES

- [1] W. Shi, J. Cao, Q. Zhang, Y. Li, and L. Xu, "Edge computing: Vision and challenges," *IEEE Internet of Things Journal*, vol. 3, no. 5, pp. 637–646, 2016.
- [2] M. Satyanarayanan, "The emergence of edge computing," *Computer*, vol. 50, no. 1, pp. 30–39, 2017.
- [3] Y. Mao, C. You, J. Zhang, K. Huang, and K. B. Letaief, "A survey on mobile edge computing: The communication perspective," *IEEE Commun. Surveys Tuts.*, vol. 19, no. 4, pp. 2322–2358, 2017.
- [4] S. Wang, R. Ugaonkar, M. Zafer, K. Chan, and K. K. Leung, "Dynamic task offloading for edge computing in IoT networks," *IEEE Access*, vol. 7, pp. 62745–62756, 2019.
- [5] H. Chen, W. Qin, and L. Wang, "Task partitioning and offloading in IoT cloud-edge collaborative computing framework: A survey," *J. Cloud Comput.*, vol. 11, no. 1, pp. 1–25, 2022.
- [6] J. Xu, L. Chen, and P. Zhou, "Joint service caching and task offloading for mobile edge computing in dense networks," in *Proc. IEEE INFOCOM*, 2018, pp. 1–9.
- [7] Z. Li and Y. Zhao, "Congestion control in internet of things (IoT) using auction theory," *Sci. Rep.*, vol. 14, no. 1, pp. 1–15, 2024.
- [8] Z. Han, D. Niyato, W. Saad, T. Başar, and A. Hjørungnes, *Game Theory in Wireless and Communication Networks*. Cambridge, U.K.: Cambridge Univ. Press, 2012.
- [9] W. Saad, Z. Han, H. V. Poor, and T. Başar, "Game-theoretic methods for the smart grid," *IEEE Signal Process. Mag.*, vol. 29, no. 5, pp. 86–105, 2014.
- [10] R. W. Rosenthal, "A class of games possessing pure-strategy Nash equilibria," *Int. J. Game Theory*, vol. 2, no. 1, pp. 65–67, 1973.
- [11] D. Monderer and L. S. Shapley, "Potential games," *Games Econom. Behav.*, vol. 14, no. 1, pp. 124–143, 1996.
- [12] Z. Niu, X. Zhang, and Y. Sun, "A game-theoretic approach for congestion-aware task scheduling in edge computing," *IEEE Trans. Mobile Comput.*, vol. 18, no. 10, pp. 2391–2404, 2019.
- [13] Y. Liang, J. Yu, and B. Chen, "Game theory-based task offloading and resource allocation in mobile edge computing," *Future Gener. Comput. Syst.*, vol. 108, pp. 24–34, 2020.
- [14] X. Pei, P. Sun, Y. Hu, D. Li, B. Chen, and L. Tian, "Edge computing offloading strategy for space-air-ground integrated network based on game theory," *Comput. Netw.*, vol. 241, art. no. 110331, 2024.
- [15] K. Jia, H. Xia, R. Zhang, Y. Sun, and K. Wang, "Multi-agent DRL for edge computing: A real-time proportional compute offloading," *Comput. Netw.*, vol. 252, art. no. 110665, 2024.
- [16] M. A. Ferrag, O. Friha, D. Hamouda, L. Maglaras, and H. Janicke, "Edge-IIoTset: A new comprehensive realistic cyber security dataset of IoT and IIoT applications for centralized and federated learning," *IEEE Access*, vol. 10, pp. 40281–40306, 2022.
- [17] T. Wang, L. Zhou, and Y. Qian, "Dynamic task offloading and online scheduling for edge-enabled IoT with a hierarchical framework," *Comput. Netw.*, vol. 269, p. 111486, 2025.
- [18] A. M. Alwakeel, "Synergistic integration of edge computing and 6G networks for real-time IoT applications," *Mathematics*, vol. 13, no. 9, art. no. 1540, 2025.
- [19] J. Zhai, J. Bi, H. Yuan, J. Zhang, and R. Buyya, "Energy-efficient and latency-aware task offloading for industrial cloud-edge systems with heterogeneous CPUs and GPUs," *IEEE Internet Things J.*, vol. 12, no. 13, p. 25757, 2025.
- [20] X. Liu, Y. Chen, and Z. Wang, "Distributed task offloading in edge computing: A multi-objective adaptive deep reinforcement learning algorithm," *Engineering Applications of Artificial Intelligence*, vol. 141, p. 110120, 2025.
- [21] Y. Zhang, H. Li, and J. Xu, "Lyapunov-guided multi-agent deep reinforcement learning for Industrial IoT task offloading," *Internet of Things*, vol. 30, p. 101604, 2025.
- [22] L. Gao, P. Sun, and K. Liu, "Graph attention reinforcement learning for congestion-aware task offloading in edge computing," *Electronics*, vol. 14, no. 2, p. 312, 2025.
- [23] J. Wang, Z. Zhao, and T. Li, "Traffic-aware intelligent association and task offloading in mobile edge computing networks," *Electronics*, vol. 13, no. 16, p. 3130, 2024.
- [24] Y. Chen, M. Zhou, and H. Xu, "A two-level game-theoretic approach for task offloading and resource allocation in mobile edge computing," *Engineering Applications of Artificial Intelligence*, vol. 135, p. 108847, 2024.